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STW7085CEM - Advanced Machine Learning
Task 2, Part 1 - Fuzzy Logic Controller
SmartHome Care Environment FLC
Fixed for MATLAB R2019b and newer

clear; clc; close all;

---- Step 1: Create Mamdani FIS

```
fis = mamfis('Name', 'SmartHome_Care_FLC');  
fis.AndMethod      = 'min';  
fis.OrMethod       = 'max';  
fis.ImplicationMethod = 'min';  
fis.AggregationMethod = 'max';  
fis.DefuzzificationMethod = 'centroid';
```

---- Step 2: Input 1 - Temperature (°C)

```
fis = addInput(fis, [10 40], 'Name', 'Temperature');
fis = addMF(fis, 'Temperature', 'trapmf', [10 10 14 18], 'Name', 'Cold');
fis = addMF(fis, 'Temperature', 'trimf', [14 18 22], 'Name', 'Cool');
fis = addMF(fis, 'Temperature', 'trimf', [19 23 26], 'Name',
'Comfortable');
fis = addMF(fis, 'Temperature', 'trimf', [23 27 31], 'Name', 'Warm');
fis = addMF(fis, 'Temperature', 'trapmf', [28 32 40 40], 'Name', 'Hot');
```

---- Step 3: Input 2 - Humidity (%)

```
fis = addInput(fis, [20 90], 'Name', 'Humidity');
fis = addMF(fis, 'Humidity', 'trapmf', [20 20 30 45], 'Name', 'Low');
fis = addMF(fis, 'Humidity', 'trimf', [30 55 70], 'Name', 'Moderate');
fis = addMF(fis, 'Humidity', 'trapmf', [60 75 90 90], 'Name', 'High');
```

---- Step 4: Input 3 - ActivityLevel (0-10)

```
fis = addInput(fis, [0 10], 'Name', 'ActivityLevel');
fis = addMF(fis, 'ActivityLevel', 'trapmf', [0 0 1 3], 'Name',
'Sleeping');
fis = addMF(fis, 'ActivityLevel', 'trimf', [1.5 4.5 7.5], 'Name',
'Resting');
fis = addMF(fis, 'ActivityLevel', 'trapmf', [6.5 9 10 10], 'Name',
'Active');
```

---- Step 5: Input 4 - AmbientLight (lux)

```
fis = addInput(fis, [0 1000], 'Name', 'AmbientLight');
fis = addMF(fis, 'AmbientLight', 'trapmf', [0 0 100 300], 'Name',
'Dark');
fis = addMF(fis, 'AmbientLight', 'trimf', [150 400 700], 'Name',
'Dim');
fis = addMF(fis, 'AmbientLight', 'trapmf', [500 800 1000 1000], 'Name',
'Bright');
```

---- Step 6: Output 1 - HVAC_Power (%)

```
fis = addOutput(fis, [-100 100], 'Name', 'HVAC_Power');
fis = addMF(fis, 'HVAC_Power', 'trapmf', [-100 -100 -70 -40], 'Name',
```

```

=====
=====
=====
'FullCool');
fis = addMF(fis, 'HVAC_Power', 'trimf', [-55 -25 0], 'Name',
'MildCool');
fis = addMF(fis, 'HVAC_Power', 'trimf', [-20 0 20], 'Name', 'Off');
fis = addMF(fis, 'HVAC_Power', 'trimf', [0 25 55], 'Name',
'MildHeat');
fis = addMF(fis, 'HVAC_Power', 'trapmf', [40 70 100 100], 'Name',
'FullHeat');

```

---- Step 7: Output 2 - LightDimmer (%)

```

fis = addOutput(fis, [0 100], 'Name', 'LightDimmer');
fis = addMF(fis, 'LightDimmer', 'trapmf', [0 0 3 12], 'Name', 'Off');
fis = addMF(fis, 'LightDimmer', 'trimf', [5 22 40], 'Name', 'Low');
fis = addMF(fis, 'LightDimmer', 'trimf', [30 50 70], 'Name', 'Medium');
fis = addMF(fis, 'LightDimmer', 'trimf', [60 78 92], 'Name', 'High');
fis = addMF(fis, 'LightDimmer', 'trapmf', [85 95 100 100], 'Name', 'Full');

```

---- Step 8: Rule Base -----

```

ruleList = [
    1 0 1 0 5 1 1 1 ;
    1 0 2 0 5 2 1 1 ;
    1 0 3 0 4 3 1 1 ;
    2 0 1 0 4 1 1 1 ;
    2 0 2 0 3 2 1 1 ;
    2 0 3 0 3 3 1 1 ;
    3 2 1 0 3 1 1 1 ;
    3 2 2 0 3 2 1 1 ;
    3 2 3 0 3 3 1 1 ;
    3 3 0 0 2 0 0.8 1 ;
    3 1 3 0 4 0 0.7 1 ;
    4 0 1 0 2 1 1 1 ;
    4 0 2 0 2 2 1 1 ;
    4 0 3 0 1 3 1 1 ;
    4 3 0 0 1 0 0.9 1 ;
    5 0 1 0 1 1 1 1 ;
    5 0 2 0 1 2 1 1 ;
    5 0 3 0 1 3 1 1 ;
    5 3 0 0 1 0 1 1 ;
    0 0 1 1 0 1 1 1 ;
    0 0 2 1 0 3 1 1 ;
    0 0 3 1 0 4 1 1 ;
    0 0 2 2 0 2 1 1 ;
    0 0 3 2 0 3 1 1 ;
    0 0 0 3 0 1 1 1 ;
];

```

```

fis = addRule(fis, ruleList);

```

---- Step 9: Save FIS -----

```
writeFIS(fis, 'SmartHome_Care_FLC');
disp('FIS saved as SmartHome_Care_FLC.fis');
```

FIS saved as SmartHome_Care_FLC.fis

---- Step 10: Test Scenarios

```
fprintf('\n== Test Scenarios ==\n');
fprintf('%-35s %-12s %-14s\n', 'Scenario', 'HVAC (%)', 'Dimmer (%)');
fprintf('%s\n', repmat('-',1,62));
```

```
scenarios = {
    'Cold night, sleeping, dark',      [12 50 0.5 50];
    'Cold morning, resting, dim',      [14 45 2 300];
    'Comfortable, active, bright day', [23 55 7 700];
    'Hot afternoon, resting, bright',  [32 60 2 900];
    'Warm evening, active, dim',       [28 50 8 400];
    'Very hot, sleeping, dark',       [36 65 0.5 20];
    'Hot & humid, resting, dim',       [31 80 3 400];
    'Cold, active, dark',             [13 40 9 50];
};
```

```
for i = 1:length(scenarios)
    name = scenarios{i,1};
    inp = scenarios{i,2};
    out = evalfis(fis, inp);
    fprintf('%-35s %-12.1f %-14.1f\n', name, out(1), out(2));
end
```

== Test Scenarios ==

Scenario	HVAC (%)	Dimmer (%)
Cold night, sleeping, dark	77.2	3.9
Cold morning, resting, dim	74.1	14.2
Comfortable, active, bright day	0.0	27.2
Hot afternoon, resting, bright	-74.1	11.5
Warm evening, active, dim	-74.7	50.0
Very hot, sleeping, dark	-77.2	3.9
Hot & humid, resting, dim	-75.7	22.4
Cold, active, dark	26.7	61.7

---- Step 11: Plot MFs -----

```
figure('Name','Input MFs','Position',[100 100 900 700]);
subplot(2,2,1); plotmf(fis,'input',1); title('Temperature (°C)');
subplot(2,2,2); plotmf(fis,'input',2); title('Humidity (%)');
subplot(2,2,3); plotmf(fis,'input',3); title('Activity Level');
```

```

=====
=====
=====
subplot(2,2,4); plotmf(fis,'input',4); title('Ambient Light (lux)');
sgtitle('Input Membership Functions');

```

```

figure('Name','Output MFs','Position',[100 450 900 350]);
subplot(1,2,1); plotmf(fis,'output',1); title('HVAC Power (%)');
subplot(1,2,2); plotmf(fis,'output',2); title('Light Dimmer (%)');
sgtitle('Output Membership Functions');

```

```

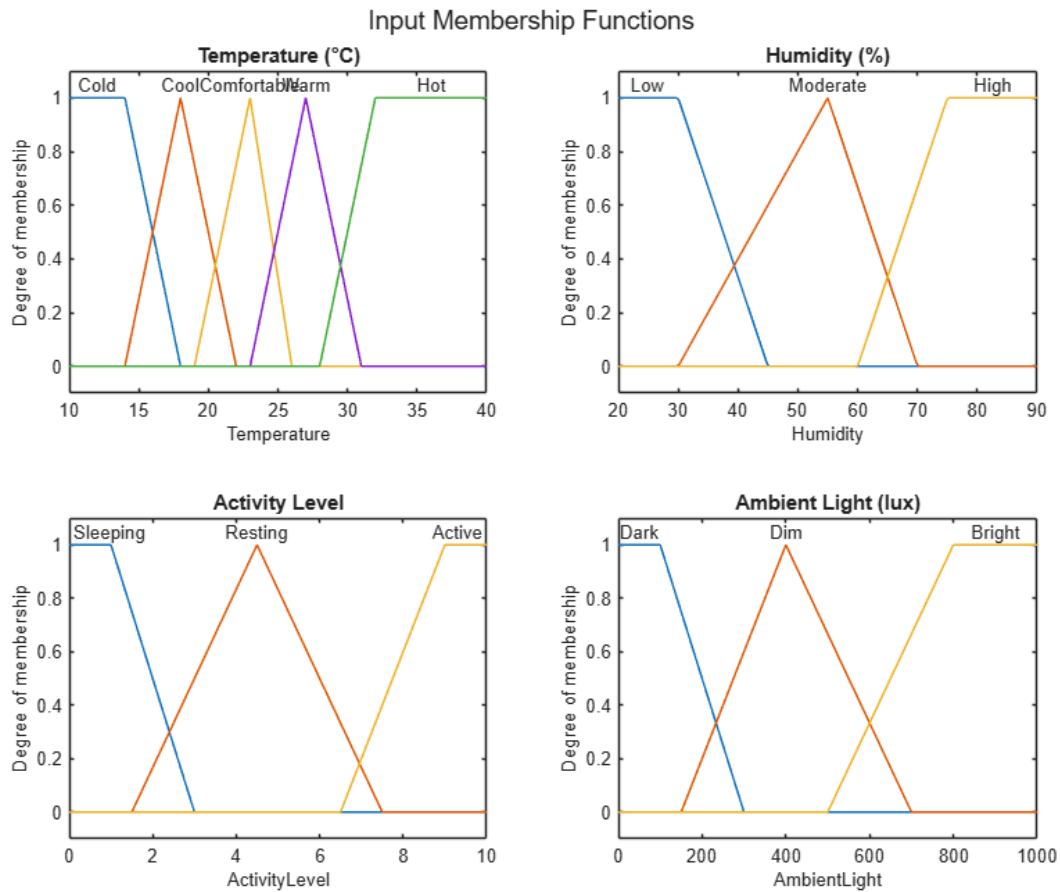
disp('=== FLC Complete ===');
disp('Open Rule Viewer : ruleview(fis)');
disp('Open Surface View: surfview(fis)');

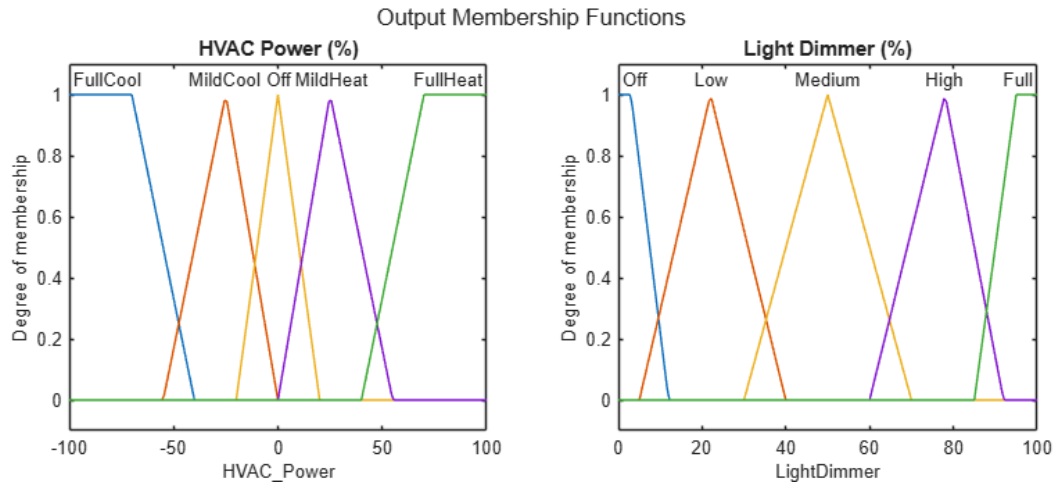
```

```

=== FLC Complete ===
Open Rule Viewer : ruleview(fis)
Open Surface View: surfview(fis)

```





---- Step 12: Display all rules as text

=====

```
fprintf('\n=== Rule Base (25 rules) ===\n');
showrule(fis);
```

```
=== Rule Base (25 rules) ===
```

---- Step 13: 3D Control Surface Plots

=====

The assignment marking scheme (7 marks) specifically asks for "control surface plots demonstrating how the controller achieves the specified behaviours in relation to an operational scenario."

Three surfaces are generated below:

- Surface A – Temperature × ActivityLevel → HVAC_Power
- Surface B – Temperature × Humidity → HVAC_Power
- Surface C – AmbientLight × ActivityLevel → LightDimmer

```
fprintf('\nGenerating control surface plots (this may take ~30 s)...\n');
```

```
figure('Name','Control Surfaces','NumberTitle','off', ...
       'Position',[100 100 1400 480]);
```

```
Generating control surface plots (this may take ~30 s)...
```

=====

=====

===

-- Surface A: Temp × Activity → HVAC

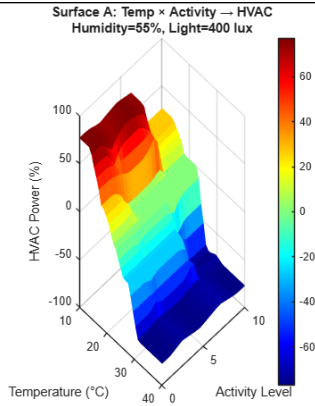
Fix: Humidity = 55 (Moderate), AmbientLight = 400 (Dim)

```
subplot(1,3,1);

T_vec = linspace(10, 40, 35);
A_vec = linspace(0, 10, 25);
[T_grid, A_grid] = meshgrid(T_vec, A_vec);
HVAC_grid = zeros(size(T_grid));

for r = 1:size(T_grid,1)
    for c = 1:size(T_grid,2)
        out = evalfis(fis, [T_grid(r,c), 55, A_grid(r,c), 400]);
        HVAC_grid(r,c) = out(1);
    end
end

surf(T_grid, A_grid, HVAC_grid, 'EdgeColor','none');
colorbar; colormap(subplot(1,3,1), jet);
xlabel('Temperature (°C)', 'FontSize', 11);
ylabel('Activity Level', 'FontSize', 11);
zlabel('HVAC Power (%)', 'FontSize', 11);
title({'Surface A: Temp × Activity → HVAC'; ...
    'Humidity=55%, Light=400 lux'}, 'FontSize', 11, 'FontWeight', 'bold');
view(45, 30); grid on;
```



-- Surface B: Temp × Humidity → HVAC

Fix: ActivityLevel = 5 (Resting), AmbientLight = 400 (Dim)

```
subplot(1,3,2);

H_vec = linspace(20, 90, 30);
[T2_grid, H_grid] = meshgrid(T_vec, H_vec);
HVAC2_grid = zeros(size(T2_grid));

for r = 1:size(T2_grid,1)
    for c = 1:size(T2_grid,2)
        out = evalfis(fis, [T2_grid(r,c), H_grid(r,c), 5, 400]);
        HVAC2_grid(r,c) = out(1);
    end
end

surf(T2_grid, H_grid, HVAC2_grid, 'EdgeColor','none');
colorbar; colormap(subplot(1,3,2), jet);
xlabel('Temperature (°C)', 'FontSize', 11);
ylabel('Humidity (%)', 'FontSize', 11);
zlabel('HVAC Power (%)', 'FontSize', 11);
title({'Surface B: Temp × Humidity → HVAC'; ...
    'Activity=5 (Resting), Light=400 lux'},
    'FontSize', 11, 'FontWeight', 'bold');
view(45, 30); grid on;
```

Warning: Unable to use defuzzification method 'centroid' on empty fuzzy set of output 1. Defuzzified output value set to its mean range value 0.

Warning: Unable to use defuzzification method 'centroid' on empty fuzzy set of output 1. Defuzzified output value set to its mean range value 0.

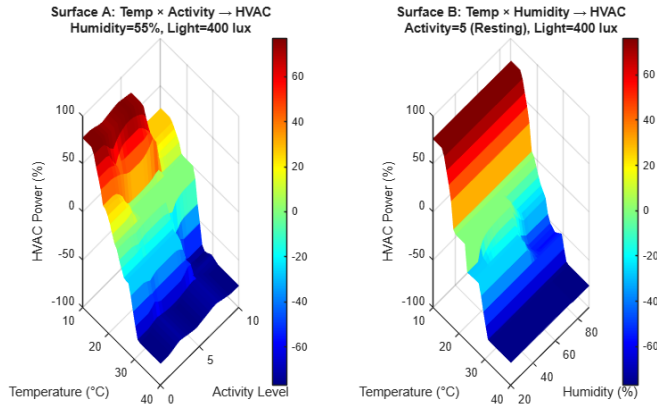
Warning: Unable to use defuzzification method 'centroid' on empty fuzzy set of output 1. Defuzzified output value set to its mean range value 0.

Warning: Unable to use defuzzification method 'centroid' on empty fuzzy set of output 1. Defuzzified output value set to its mean range value 0.

```

of
output 1. Defuzzified output value set to its mean range value 0.
Warning: Unable to use defuzzification method 'centroid' on empty fuzzy set
of
output 1. Defuzzified output value set to its mean range value 0.

```



-- Surface C: AmbientLight × Activity → Light-Dimmer -----

Fix: Temperature = 23 (Comfortable), Humidity = 55 (Moderate)

```

subplot(1,3,3);

L_vec = linspace(0, 1000, 35);
[L_grid, A2_grid] = meshgrid(L_vec, A_vec);
Dim_grid = zeros(size(L_grid));

for r = 1:size(L_grid,1)
    for c = 1:size(L_grid,2)
        out = evalfis(fis, [23, 55, A2_grid(r,c), L_grid(r,c)]);
        Dim_grid(r,c) = out(2);
    end
end

surf(L_grid, A2_grid, Dim_grid, 'EdgeColor','none');
colorbar; colormap(subplot(1,3,3), parula);
xlabel('Ambient Light (lux)','FontSize',11);
ylabel('Activity Level','FontSize',11);
zlabel('Dimmer (%)','FontSize',11);
title({'Surface C: Light × Activity → Dimmer'; ...
    'Temp=23°C, Humidity=55%'}, 'FontSize',11,'FontWeight','bold');
view(225, 30); grid on;

sgtitle('SmartHome FLC – Control Surface Plots (Part 1, 7-mark criterion)',
...
    'FontSize',13,'FontWeight','bold');

fprintf('Control surface plots complete.\n');

```

```

=====
=====
===
fprintf('\n== Operational Scenario Analysis ==\n');
fprintf('Surface A shows HVAC output decreasing (heating → cooling) as\n');
fprintf('  temperature rises from Cold to Hot across all activity\n');
fprintf('  levels.\n');
fprintf('  High activity (Active) triggers stronger cooling at any\n');
fprintf('  temperature.\n');
fprintf('  due to body heat generation – visible as downward shift at\n');
fprintf('  Act=8-10.\n\n');
fprintf('Surface B confirms humidity-aware cooling: at Comfortable\n');
fprintf('  temperature\n');
fprintf('  (23°C), HVAC shifts from 0 to MildCool (-25%) as Humidity\n');
fprintf('  rises\n');
fprintf('  above 65% (rule R10, weight=0.8), preventing perceived\n');
fprintf('  discomfort.\n\n');
fprintf('Surface C shows the inverse light-dimmer relationship: dimmer\n');
fprintf('  output\n');
fprintf('  approaches 0 as AmbientLight exceeds 650 lux regardless of\n');
fprintf('  activity.\n');
fprintf('  In Dark (<300 lux) conditions, Active users receive up to 80%\n');
fprintf('  \n');
fprintf('  dimmer level while Sleeping users always get 0% (rule R20).\n');

```

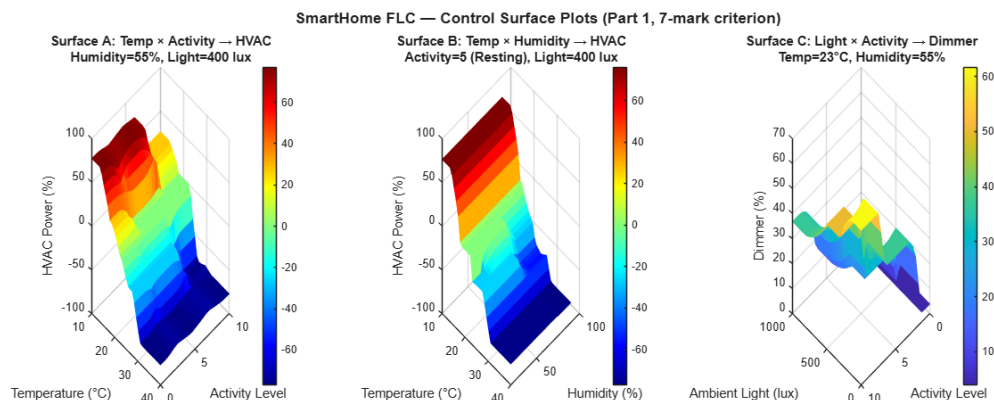
Control surface plots complete.

=== Operational Scenario Analysis ===

Surface A shows HVAC output decreasing (heating → cooling) as temperature rises from Cold to Hot across all activity levels. High activity (Active) triggers stronger cooling at any temperature due to body heat generation – visible as downward shift at Act=8-10.

Surface B confirms humidity-aware cooling: at Comfortable temperature (23°C), HVAC shifts from 0 to MildCool (-25%) as Humidity rises above 65% (rule R10, weight=0.8), preventing perceived discomfort.

Surface C shows the inverse light-dimmer relationship: dimmer output approaches 0 as AmbientLight exceeds 650 lux regardless of activity. In Dark (<300 lux) conditions, Active users receive up to 80% dimmer level while Sleeping users always get 0% (rule R20).



=====

=====

===

---- Step 14: Rule Viewer reminder

=====

```
fprintf('\n--- To view rule activation for a specific input ---\n');
fprintf('Run in MATLAB command window:\n');
fprintf('  ruleview(fis)\n');
fprintf('  Then set inputs e.g. [30, 60, 8, 200] to see rules fire for:\n');
fprintf('  Temp=30°C, Hum=60%, Active=8, Light=200lux\n');
```

--- To view rule activation for a specific input ---

Run in MATLAB command window:

ruleview(fis)

Then set inputs e.g. [30, 60, 8, 200] to see rules fire for:

Temp=30°C, Hum=60%, Active=8, Light=200lux

Published with MATLAB® R2026a